

Also, we disregard all those questions related to integral points in convex bodies — this leads to the beautiful theory that was named the “geometry of numbers” by its founder, Hermann Minkowski [407]. Modern treatments are Cassels [143] and Gruber & Lekkerkerker [250]. The algorithmic questions are treated in Kannan [310], Lagarias [346], and Schrijver [484]. See also Erdős, Gruber & Hammer [199] for a nice “problem-oriented” survey.

Furthermore, we do not have the time or space to treat much more of the aspects of linear and integral optimization related to convex polytopes. Besides traveling salesman polytopes, many other classes have been studied extensively. It seems that cut polytopes are especially important for practical applications — see Deza & Laurent [185].

The necessity to optimize over polytopes with only partial information about their facets leads to “cutting plane algorithms”: the books by Schrijver [484] and by Grötschel, Lovász & Schrijver [246] explain the powerful theory behind this. Two recent references that describe the method for “how to find a good solution for a Traveling Salesman Problem if you really need one” are Reinelt [452] and Jünger, Reinelt & Thienel [296]. The “New York Times” and “New Scientist” articles [423] [340], and the survey by Grötschel & Padberg [248], are references for the spectacular success of the method on extremely large traveling salesman problems. Further success in the race for the “TSP Olympics” (i.e., for “largest traveling salesman problem ever solved”) was reported in [21]: David Applegate, Bob Bixby, Vašek Chvátal and Bill Cook have been able to solve a 13,509-city instance to optimality, using a polyhedral approach, LP-relaxations, a branch&cut framework, very clever heuristics, superior programming, and a network of 48 powerful workstations. Finally, in May 2004 the same authors together with Keld Helsgaun solved a 24,978 city instance to optimality [22] [23].

Problems and Exercises

0.0 Given a 3-dimensional polytope such that every two vertices are adjacent, show that it is a tetrahedron.

0.1 Show that if a polytope is both simple and simplicial, then it is a simplex or an n -gon.

Similarly, if a d -polytope is simple and *cubical* (i.e., all its facets are combinatorially equivalent to $(d - 1)$ -cubes), then P is a d -cube or an n -gon.

0.2 Prove that a polytope can be represented as the affine image of a crosspolytope if and only if it is centrally symmetric.

Show that if P is a zonotope (an affine image of a d -cube), then every face of P is centrally symmetric as well. What about the converse?

- 0.3 Show that the permutahedron $\Pi_{d-1} \subseteq \mathbb{R}^d$ (Example 0.10) has dimension $d - 1$, that it is a zonotope, and that it is simple.

Describe its $2^d - 2$ facets, by constructing inequalities that determine them.

- 0.4 Let $a_1 \geq a_2 \geq \dots \geq a_d$ be real numbers, not all equal. The *generalized permutahedron* (or *orbit polytope*) $\Pi_{d-1}(a_1, \dots, a_d)$ is the convex hull of all the vectors given by all the permutations of the multiset $\{a_1, \dots, a_d\}$.

Investigate the combinatorics of the generalized permutahedra. In particular, show that their dimension is $d - 1$. Are they all simple? (They are not.)

Under what conditions do all the edges of $\Pi_{d-1}(a_1, \dots, a_n)$ have the same length? (Schoute [481, p. 5])

- 0.5 Let $P = C_d \subseteq \mathbb{R}^d$ be the d -cube. Enumerate the $3^d + 1$ faces of C_d , and show that the nonempty faces are naturally associated with the sign vectors in $\{+, -, 0\}^d$.

Given a linear function $\mathbf{c} \in (\mathbb{R}^d)^*$, how can one find a vertex that maximizes $\mathbf{c}$ over P (“optimization problem”)?

Given $\mathbf{y} \in \mathbb{R}^d$, how do we tell whether $\mathbf{y} \in P$? If $\mathbf{y} \notin P$, how can we find an inequality that is valid for P but is violated by $\mathbf{y}$ (“separation problem”)?

For which other classes of polytopes discussed in Lecture 0 can you easily solve these problems?

- 0.6 Describe $C_d(d + 2)$, the cyclic d -polytopes with $d + 2$ vertices, combinatorially and explicitly.

Is the 2-neighborly polytope $(\Delta_2 \times \Delta_2)^\Delta$ constructed in Example 0.5 combinatorially equivalent to $C_4(6)$?

- 0.7 Consider the cyclic polytope $C_d(n) = \text{conv}\{\mathbf{x}(0), \mathbf{x}(2), \dots, \mathbf{x}(n-1)\}$. Show that there is an affine symmetry (an affine reflection) which induces the symmetry $i \longleftrightarrow n + 1 - i$ (that is, $\mathbf{x}(i-1) \longleftrightarrow \mathbf{x}(n-i)$), and thus the corresponding combinatorial symmetry of $C_d(n)$.

- 0.8 From Gale’s evenness condition, given in Theorem 0.7, derive a complete combinatorial description of all the faces of $C_d(n)$. From this, derive that the cyclic polytopes are $\lfloor \frac{d}{2} \rfloor$ -neighborly (Corollary 0.8).

- 0.9 Show (bijectively) that the number of ways in which $2k$ elements can be chosen from $[n]$ in “even blocks of adjacent elements” is $\binom{n-k}{k}$.

Thus, derive from Gale’s evenness condition that the formula for the number of facets of $C_d(n)$ is

$$f_{d-1}(C_d(n)) = \binom{n - \lceil \frac{d}{2} \rceil}{\lfloor \frac{d}{2} \rfloor} + \binom{n - 1 - \lceil \frac{d-1}{2} \rceil}{\lfloor \frac{d-1}{2} \rfloor},$$

where $\lceil \cdot \rceil$ is the round-up function, with $\lceil \frac{k}{2} \rceil = k - \lfloor \frac{k}{2} \rfloor$. Here the first term corresponds to the facets for which the first block is even, and the second term corresponds to the cases where the first block is odd. Deduce

$$f_{d-1}(C_d(n)) = \begin{cases} \frac{n}{n-k} \binom{n-k}{k} & \text{for } d = 2k \text{ even,} \\ 2 \binom{n-k-1}{k} & \text{for } d = 2k + 1 \text{ odd.} \end{cases}$$

How many facets do the cyclic polytopes $C_{10}(20)$, $C_{10}(100)$, and $C_{50}(100)$ have, approximately?

- 0.10 Show that if a polytope is k -neighborly, then every $(2k-1)$ -face is a simplex. Conclude that if a d -polytope is $(\lfloor \frac{d}{2} \rfloor + 1)$ -neighborly, then it is a simplex.
- 0.11* Is there a fast and simple way to decide whether a certain point $\mathbf{x} \in \mathbb{R}^d$ (with rational coordinates, say) is contained in the cyclic polytope $C_d(1, 2, \dots, n)$?
(General theory — namely the polynomial equivalence of optimization and separation according to Grötschel, Lovász & Schrijver [246] — implies that there is a polynomial algorithm for this task, since optimization over $C_d(n)$ is easy, by comparing the vertices. However, we ask for a simple combinatorial test, not using the ellipsoid method.)
- 0.12 Prove the claims in Example 0.12 about the Birkhoff polytope $P(d)$: in particular, show that the dimension is $(d-1)^2$, and that the number of facets is d^2 .
The Birkhoff polytope $P(d)$ and the permutahedron Π_{d-1} are closely related: show that there is a canonical projection map $P(d) \longrightarrow \Pi_{d-1}$.
- 0.13 Draw the 3-dimensional associahedron K_3 . Justify the general formula $\frac{1}{n} \binom{2n-2}{n-1}$ for the number of vertices of K_{n-2} .
- 0.14 Describe the combinatorial structure of the traveling salesman polytopes $Q_T(3)$, $Q_T(4)$, and $Q_T(5)$. How many vertices and facets do they have? Which vertices are adjacent? Are they simple, or simplicial? Similarly, try to describe $Q'_T(2)$, $Q'_T(3)$, and $Q'_T(4)$.

- 0.15* What is the maximal number $f(d)$ of facets of a d -dimensional 0/1-polytope? How fast does $f(d)$ grow asymptotically? (It is not hard to see that

$$2^d \leq f(d) \leq d! + 2d.$$

The upper bound was suggested by Imre Bárány, via the following observation: if we add the “missing vertices” of the d -cube to the polytope one by one, then we add a volume of at least $1/d!$ for every facet that is destroyed of the original 0/1-polytope. The process will stop with the d -cube of volume 1, with only $2d$ facets.

More recent progress on this problem is recorded in Kortenkamp, Richter-Gebert, Sarangarajan & Ziegler [343]. Still, there is a huge gap between the lower and the upper bounds. We know

$$\begin{aligned} f(1) = 2, \quad f(2) = 4, \quad f(3) = 8, \quad f(4) = 16, \\ f(5) = 40, \quad 121 \leq f(6) \leq 610, \quad \text{etc.} \end{aligned}$$

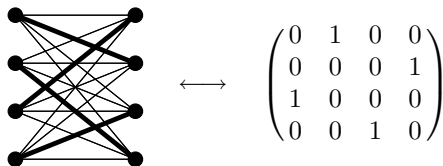
for small dimensions. (See Aichholzer [5, 6] for enumeration techniques.) Asymptotically the best known bounds are

$$(3.6)^d < f(d) \leq 30(d-2)!$$

for all large enough d . Here the upper bound is due to Fleiner, Kaibel & Rote [206], while the lower bound is from explicit computation of “random 0/1-polytopes” in low dimensions in combination with a “free sum” construction for 0/1-polytopes from [343]. The value 3.6 was achieved in March 1997 by Thomas Christof for a random 0/1-polytope (of dimension 13, with 254 vertices and *at least* 17,464,356 facets), using his PORTA code and new ideas described in Christof & Reinelt [155].

For “current records” in the “Olympic race” for 0/1-polytopes with many facets see [342] on the Web.)

- 0.16 Via the construction of “characteristic vectors” from Example 0.13, show that the vertices of the Birkhoff polytope $P(n)$ correspond to the perfect matchings in the complete bipartite graph $K_{n,n}$.



- 0.17 Show that the Birkhoff polytope $P(d) \subseteq \mathbb{R}^{d^2}$ contains the asymmetric traveling salesman polytope $Q_T(d) \subseteq \mathbb{R}^{d^2-d} \subseteq \mathbb{R}^{d^2}$. Does every facet of $P(d)$ yield a facet of $Q_T(d)$? (For a detailed investigation, see Billera & Sarangarajan [76].)

1

Polytopes, Polyhedra, and Cones

In this lecture we prove some fundamental properties, in particular the equivalence of the two definitions of polytopes in Definition 0.1.

Of course, one could ask whether it is really necessary to go through these details, since the result is quite obvious anyway, and complete proofs are in the books [133] [252] [403] [484]. There are several good reasons. One is that we can give proofs that introduce important machinery (like Fourier-Motzkin elimination), which is useful for other purposes as well. It also yields a basic algorithmic tool to deal with polytopes. Additionally, these proofs provide geometric intuition, which we will need later. We will also see polarity appear in this context quite naturally, because we do two versions of Fourier-Motzkin, which are related by polarity. The “usual” approach is to do only one version, and prove the second half using polarity — this saves some work, but avoids the very interesting polar version. Finally, our proofs are (meant to be) easy and transparent, following simple geometric ideas through some elementary linear algebra, so they might even be fun. (There should be no crying in this lecture.)

1.1 The “Main Theorem”

However, to make sure that the pain level does not go below zero, we start with a few definitions. In the following, we work with two versions of polyhedra — in the course of this lecture we will see that they are mathematically (but not algorithmically!) equivalent. The two concepts